

Similar Triangles

DSAT Math Practice | 25 Questions | Difficulty-Graded | ~15-20% Grid-Ins

Directions: Answer each question by selecting the best answer from the choices given, or by entering your answer in the grid-in box for student-produced response questions. You may use a calculator on all questions. Reference the provided formulas as needed.

1

Easy

Triangles JKL and MNP are similar. In triangle JKL, $JK = 6$ and $KL = 9$. In triangle MNP, $MN = 10$. What is the length of NP?

A

12

B

15

C

18

D

20

2

Easy

Two similar triangles have a scale factor of $3 : 5$. If the shortest side of the smaller triangle is 12 centimeters, what is the length, in centimeters, of the shortest side of the larger triangle?

A

7.2

B

15

C

20

D

36

3

Easy

In the figure, line segment DE is parallel to line segment BC. If $AD = 4$, $DB = 6$, and $DE = 5$, what is the length of BC?

A

7.5

B

8

C

12.5

4

Easy

Triangle RST is similar to triangle UVW, where R corresponds to U and S corresponds to V. If $RS = 8$, $ST = 14$, and $UV = 12$, what is the length of VW?

A

16

B

18

C

20

D

21

5

Easy

Triangles ABC and DEF are similar, with AB corresponding to DE. If $AB = 15$, $BC = 20$, and $DE = 9$, what is the length of EF?

[Student-Produced Response]

6

Medium

In right triangle PQR, altitude QS is drawn from the right angle at Q to hypotenuse PR. If $PS = 4$ and $SR = 9$, what is the length of QS?

A

5

B

6

C

6.5

D

13

7

Medium

A vertical flagpole casts a shadow 18 feet long at the same time a nearby 5-foot-tall fence post casts a shadow 6 feet long. What is the height, in feet, of the flagpole?

A

12

B

15

- C 21.6
- D 54

8

Medium

In the xy -plane, triangle ABC has vertices $A(0, 0)$, $B(8, 0)$, and $C(0, 6)$. Triangle DEF is similar to triangle ABC, with a scale factor of $\frac{5}{2}$. What is the length of the side in triangle DEF that corresponds to side AC?

- A 2.4
- B 10
- C 12
- D 15

9

Medium

Two similar triangles have perimeters of 36 and 54. If the longest side of the smaller triangle is 15, what is the longest side of the larger triangle?

- A 10
- B 18
- C 22.5
- D 25

10

Medium

Line segment FG is parallel to line segment HI in triangle EHI. If $EF = 7$, $FH = 3$, and $EG = 14$, what is the length of GI?

[Student-Produced Response]

11

Medium

Triangles WXY and WZA are similar. $WX = 10$, $WY = 16$, and $WZ = 15$. What is the length of WA?

- A 18

- B 21
- C 24
- D 25

12**Medium**

The ratio of corresponding sides of two similar triangles is 4 : 7. If the area of the smaller triangle is 48 square inches, what is the area, in square inches, of the larger triangle?

- A 84
- B 112
- C 147
- D 196

13**Medium**

In the figure, triangles PQR and PST share angle P, and angle PQR is congruent to angle PST. If $PQ = 12$, $PS = 8$, and $PR = 15$, what is the length of PT ?

- A 8
- B 10
- C 12
- D 22.5

14**Medium**

A photographer stands 24 feet from a building. Her camera, at a height of 5 feet, is aimed at the top of the building. If the building is 53 feet tall, which expression gives the distance, in feet, from the camera to the top of the building?

- A $\sqrt{24^2 + 48^2}$
 - B $\sqrt{24^2 + 53^2}$
 - C $\sqrt{24^2 + 58^2}$
 - D $\sqrt{29^2 + 48^2}$
-

15

Medium

Triangle GHI is similar to triangle JKL. The perimeter of triangle GHI is 42 and the perimeter of triangle JKL is 63. If $GH = 10$, what is the length of JK?

[Student-Produced Response]

16

Hard

In triangle PQR, angle $P = 90$ degrees. Point S is on segment QR such that PS is perpendicular to QR. If $QS = a$ and $SR = b$, which of the following is equal to PS?

A

$(a + b)/2$

B

$\sqrt{ab}$

C

$\sqrt{a^2 + b^2}$

D

$ab/(a + b)$

17

Hard

Triangles ABC and DEF are similar with a scale factor of k (where $k > 1$), with triangle DEF being larger. If the perimeter of triangle ABC is P and the area of triangle ABC is A , which of the following gives the area of triangle DEF?

A

kA

B

k^2A

C

kP

D

k^2P

18

Hard

In the figure, BD is parallel to AE. $BC = 8$, $CD = 6$, and $CE = 15$. What is the length of AC?

A

10

B

12

C

18

D

20

19

Hard

In triangle XYZ, point M is on XY and point N is on XZ such that MN is parallel to YZ. If $\frac{XM}{MY} = \frac{3}{4}$, what is the ratio of the area of triangle XMN to the area of trapezoid MNZY?

A

 $\frac{3}{4}$

B

 $\frac{9}{16}$

C

 $\frac{9}{40}$

D

 $\frac{9}{49}$

20

Hard

Two similar right triangles have hypotenuses of length 13 and 39. If one leg of the smaller triangle is 5, what is the length of the longer leg of the larger triangle?

[Student-Produced Response]

21

Hard

In the coordinate plane, triangle ABC has vertices A(0, 0), B(10, 0), and C(4, 8). A dilation centered at the origin with scale factor $\frac{1}{2}$ maps triangle ABC to triangle A'B'C'. What is the perimeter of triangle A'B'C'?

A

 $10 + 2\sqrt{5}$

B

 $10 + 4\sqrt{5}$

C

 $20 + 4\sqrt{5}$

D

 $5 + 2\sqrt{5}$

22

Hard

In triangle TVW, line segment XY is drawn parallel to VW with X on TV and Y on TW. If TX = 5, XV = 3, and the area of triangle TXY is 50, what is the area of triangle TVW?

A

80

B

100

C

128

23

Hard

Triangle ABC is similar to triangle DEF, with ABC being the smaller triangle. The scale factor from ABC to DEF is 3 : 5. If the area of triangle ABC is 36, what is the area of triangle DEF?

[Student-Produced Response]

24

Hard

Triangle PQR has $PQ = 12$, $QR = 16$, and $PR = 20$. Triangle STU is similar to triangle PQR. If the area of triangle STU is 150, what is the perimeter of triangle STU?

A

36

B

48

C

60

D

72

25

Hard

A surveyor wants to find the width of a river. She sets up similar triangles and measures the following: the distance from her position to a point directly across the river (along the bank) is 40 meters, a reference stake 8 meters from her along the bank, and the perpendicular distance from the stake to a line of sight across the river is 15 meters. Using similar triangles, what is the width of the river, in meters?

A

45

B

60

C

75

D

120